Useful Delivery Before Capability Investment

Discover the work. Build the advantage.

A bounded portfolio study of API response assurance and transit-feed qualification

Research date: 26 October 2026. **System base:** AGI-Alpha-Agent-v0, commit `882456a`.

Status: Reproducible public-data research with useful scoped outputs. No customer purchase, live ENS validation, contract settlement, or realized commercial return is asserted.

Abstract

We searched six materially different job families and executed two scoped offerings before writing the paper. A REM transit-feed packet passes canonical structural validation and 86 independent reconciliation checks, while retaining 1,565 warnings. Three new API release questions are accepted in all four compared approaches; three additional planned questions remain source-acquisition failures. The retained conventional workflow is faster than the Workbench full index, incremental registry and graph composition. Two competent recipients reconstruct equivalent methods before receiving the asset. The original transfer control is content-identical; a separately frozen changed-content follow-up is accepted by the narrow system workflow but stopped by the second recipient's stricter input validator.

A source-frozen finite-cash simulation uses these measurements and explicitly assumed demand, attention, competition and cost conditions. Across 384 paired worlds and 1,536 policy histories, none produces positive ordinary operating surplus or funds a capability investment. The most favorable cell averages a $391.55 operating loss and a $515.97 loss after its modeled opening costs. Actual collections are zero and complete actual costs remain unknown. The supported result is useful scoped delivery with preserved reusable engineering and a negative test of the declared commercial cycle, not automation savings established against timed humans, scarce transferable capability, graph advantage, modeled self-financing succession or realized commercial success.

Decision supported by the engineering evidence. Retain the useful delivery artifacts and the simpler conventional workflow. Do not finance the tested Workbench index or graph composition on an unmeasured scarcity premium. A future investment must first establish a buyer-relevant gap that competent existing tools do not already close.

Authorship and review. AI-assisted research and engineering coordinated through AGI ALPHA issue #4850. Separate same-session agents constructed comparators, reviewed engineering, and audited economics. This is not independent external peer review. No transit publisher, Docker maintainer, or buyer has endorsed the service.

Evidence companion. Frozen source, original public inputs and licenses, accepted outputs, failed trials, process receipts, transfer records, the finite-cash simulator and its ledgers, and an offline English/French demonstration accompany this paper.

Contents

1 The customer work and the result **3**

1.1 A consequential scope boundary 4

2 Discovery before commitment **4**

2.1 Budget, selection and demand evidence 4

2.2 Why the second finalist was not expanded 5

3 What was constructed and what was already available **5**

3.1 Existing system integration 6

4 Comparisons that answer different questions **6**

4.1 Freeze, confirmation and acceptance 7

4.2 Transfer to competent recipients 8

5 Economic conditions, accounting and investment **9**

5.1 Separate cost books 9

5.2 Frozen buyer-and-competitor experiment 10

5.3 The ordinary service fails before the investment loop 11

5.4 Cheap computation is not a complete service margin 13

5.5 The proposed upgrades also lack a repayment case 13

5.6 What an investment must repay 14

6 Authenticated admission without invented authority **14**

7 Relation to the continuing research **14**

8 Limitations and the concrete missing condition **15**

9 Conclusion **15**

A Reproduction and evidence map **16**

B English/French demonstration **17**

1 The customer work and the result

The commercial hypothesis was a connected sequence: useful work, accepted efficient delivery, collected surplus, an affordable capability investment, and better fresh work. The study did not treat an implementation, a signed fixture, a benchmark count, or avoided labor as revenue. It first screened six different job families and executed two finalists using the existing SUCCESSOR Workbench where its actual extension interface permitted it.

The first offering delivers an exhaustive, source-bound inventory of declared GET 2xx JSON schema facts within a supported grammar, between two API specifications. It supports a precise review task: identify the response fields and constraints that changed, trace each change back to its source, and decide which client checks require attention. It does *not* establish live-client compatibility. The second offering delivers a REM GTFS ingestion packet for a specified week: canonical-validator results, route/date trip counts, independent reconciliation, preserved warnings and an explicit structural disposition.

These outputs have identifiable beneficiaries: integration maintainers and transit-data ingestion teams. They use real public source artifacts and support concrete review or ingestion actions. They were not commissioned by those beneficiaries. Their usefulness is supported by the completed artifacts and technical acceptance, while their willingness to pay remains unobserved. The API dossier is narrower than the initially envisioned consumer migration packet; the later source-bound consumer appendix makes the remaining gap explicit.

Table 1: All six locked API questions, including blocked deliveries. Counts are structural facts, not customer incidents.

Question	Old / new nodes	Changes	Four-arm disposition
v20.10.0 → v23.0.0	2,233 / 2,341	489	4 / 4 scoped pass
v23.0.0 → v24.0.0	2,341 / 2,343	8	4 / 4 scoped pass
v24.0.0 → v25.0.0	2,343 / 2,408	70	4 / 4 scoped pass
v26.0.0 → v27.0.0	–	–	Source path 404; 0 / 4 delivered
v27.0.0 → v28.0.0	–	–	Source path 404; 0 / 4 delivered
v28.0.0 → v29.0.0	–	–	Source path 404; 0 / 4 delivered

The primary delivered three distinct dossiers, accepted in all four arms: 12 accepted arm/question outputs out of 24 planned. Three other questions were blocked by two locked source-path 404s. On available inputs, conditional technical acceptance was 12/12. The intent-to-deliver fraction was 12/24. These denominators answer different questions and are both retained.

The REM packet covers 26 October–1 November 2026, in GTFS local service dates. Its original feed version is 20260520, with declared coverage through 31 December 2026. It contains three routes, 1,081 trip templates and 19,069 stop-time rows. The supported calendar expansion yields 443 scheduled trip instances on each weekday, 323 on Saturday and 315 on Sunday: 2,853 in total. There are 21 route/date cells and seven daily groups in one delivery.

The canonical MobilityData validator 8.0.1 reports zero errors, 1,565 warnings and one informational notice. Eighty-six independent SQLite reconciliation checks pass. The disposition is **SCOPED STRUCTURAL PASS**, with warnings retained for interpretation. This permits structural ingestion of the supplied snapshot within the declared scope; it does not certify real-time service or independently verify the timetable against actual operations. The external Java/Python stages and actual Workbench aggregation have 21.558 measured command wall seconds and 23.718 CPU seconds in total, excluding unmetered authoring and attention. The Workbench aggregation command contributes 0.115 wall

seconds and 0.123 CPU seconds.

1.1 A consequential scope boundary

The consumer review considered legacy Docker Compose 1.29.2 and Docker Engine 26. Official Docker documentation describes the omission of `Container` and `ContainerConfig` from image inspection at API version 1.45 and later [2]. The archived v25 and v26 specifications used during construction still contain these fields. A structural specification diff therefore cannot certify this behavior change or claim to have detected it. The accompanying consumer appendix binds the authentic consumer location, the archived schema evidence and the official documentation. In the previous-container path of Compose 1.29.2, line 1579 directly indexes `ContainerConfig` [3]. The unmodified, AST-extracted function returns an empty binding result for the present-key fixture and raises `KeyError` for the omitted-key fixture even when `Config` is present. Its conditional decision is to require a maintained consumer and staging evidence when this path uses API 1.45 or later. Static and extracted-function evidence are labeled as such; no live Docker service or full application migration was executed for this claim.

This boundary is economically relevant. A low-cost schema report is not equivalent to a finished migration assurance service. The modeled offering, human comparator and competitors all use the same limited dossier scope. Broader assurance would require additional implementation, verification and residual attention, whose costs cannot be erased by changing the product label.

2 Discovery before commitment

2.1 Budget, selection and demand evidence

The declared budget was six job families, no more than two development prototypes, two capability candidates, a 35-minute discovery window, a 45-minute window for each prototype, and 60 seconds per production process. These are bounded research allowances, not dollar budgets or metered human work. Agent intervals overlap and are never summed into a labor invoice. Finalists were selected from primary-source evidence and feasibility before confirmatory outcomes. Selection and later source freezes are retained separately.

API releases are authentic recurring triggers: GitHub explicitly directs maintainers to review breaking changes, update integrations and test them [1]. Transit publication is also recurrent: TransLink describes weekly feed updates and at least four seasonal changes per year, while Caltrans describes interpreting feed-quality evidence with providers [7, 8]. These facts establish recurring work classes, not sales counts for this enterprise. The selected execution feed was REM under its embedded CC BY 4.0 license, not TransLink data with additional commercial-use conditions.

The excluded families were not demonstrated to be commercially absent. SQL repair lacked a justified local-scale reproducer within the budget; accessibility had a difficult assessment boundary; numerical reference testing lacked a specific buyer job; generic migration had strong existing tooling. W3C's requirement for knowledgeable human evaluation and NIST's reference-data purpose informed those choices [11, 12]. The complete matrix preserves buyers, triggers, human workflows, alternatives, demand evidence, automation bottlenecks and uncertainty for every candidate.

Table 2: Portfolio ranked before finalist outcomes. Ordinal judgments are researcher estimates.

Rank	Family / beneficiary	Complete action and existing alternative	Selection judgment
1	API assurance / integration owners	Source-bound upgrade review; oasdiff and competent developers already exist.	Selected: public versioned inputs and exact acceptance are feasible; consumer completeness and differentiation are uncertain.
2	GTFS assurance / feed ingestion teams	Scoped ingestion disposition and service-date reconciliation; canonical MobilityData validator.	Selected: recurring schedules, reusable integration, permissive input available; no paid demand observed.
3	SQL performance repair / analytics maintainers	Equivalent query or resource fix; DuckDB optimizer and expert tuning.	Authentic issue #22474, but its large reproducer exceeds the local screen budget.
4	Accessibility repair / website maintainers	Scoped repair plus task assessment; automated tools and knowledgeable evaluators.	Whole-site acceptance cannot honestly be reduced to an automated score.
5	Numerical qualification / scientific users	Accuracy dossier tied to a real downstream decision; mature solvers and NIST reference data.	Reference cases alone are a benchmark, and the active numerical lane was excluded.
6	Generic source migration / library maintainers	Patch with regression tests; Ruff NPY201 and existing codemods.	Strong public alternatives and overlap with dependency work reduce expected additional value.

2.2 Why the second finalist was not expanded

The REM packet is a complete exploratory delivery, but this run did not establish a scarce GTFS skill, a second fresh feed comparison, a measured human-only advantage, or an economical transfer. Conventional validation was already strong. Expanding this candidate into a second unvalidated market model would have overstated the evidence. It remains a useful retained composition and a future candidate if actual recurring ingestion requirements justify it. The API family supplied the bounded repeated comparison, not a presumed commercial winner.

3 What was constructed and what was already available

The starting system had public-data Workbench adapters, canonical mission and Jobs draft compilation, evidence hashing, prior graph/authority fixtures, and general-purpose programming tools. It did not contain this source-bound API dossier adapter. That absence did not imply that other agents lacked the capability: two separate agents received reasonable search, conventional tools and reconstruction budgets, and both built equivalent analyzers.

The first retained asset is a deterministic JavaScript response index and dossier adapter. It enumerates successful GET JSON response schemas, local references, properties, arrays, additional-property schemas and structural composition branches. Facts, source pointers and reference chains remain separate. Unsupported constructs are disclosed. The second asset adds dependency-aware reuse of source-bound response subtrees. It invalidates affected roots when transitive source dependencies change. It retains a private current-session authenticity check for issued indexes; imported or tampered caches do not silently acquire trust.

Important corrections were required during construction: complete transitive source binding, explicit unsupported-keyword reporting, ownership of asynchronous inputs, private cache-chain trust, refusal to overwrite output directories, and safe UTF-16 fragment boundaries. Original source snapshots and failed controls remain in the companion. These are validated engineering improvements. They are not claims of a newly discovered indexing algorithm.

The exact contract required executable construction and checking of version binding, fragment reconstruction, evidence linkage and independent replay. The quick independent reconstructions do not establish a scarce capability or superiority over ordinary agent/tool workflows. Established public tools, however, already perform substantial API analysis. The study therefore compares against a retained conventional builder and a pinned oasdiff supplement, not against an unprepared agent with an empty skill directory.

3.1 Existing system integration

API computation runs as an actual pure-JavaScript extension of the existing Workbench. The adapter returns source-bound artifacts, and the existing engine compiles seven unfunded Jobs drafts, checks the pack, and preserves public-local execution scope. The GTFS stage is different: an external Java validator and Python/SQLite analysis precede an actual Workbench aggregation of the 21 route/date rows into seven daily totals. No arbitrary native Python dispatcher is implied.

These concrete integration points are the operational use of the existing TRANSFORMATIONAL INTELLIGENCE ∞ and SUCCESSOR Ω foundations in this study. The four reviewed repositories retain their roles. AGI-Fleet-Linux remains a deployment and fleet foundation; AGIJobsv0 and AGIJobManager retain their existing Jobs contracts, governance and settlement boundaries. This study does not establish installed four-product qualification. It does not convert the system's unfunded drafts into operational jobs, approve execution, or move funds.

4 Comparisons that answer different questions

Table 3: Four distinct estimands. An answer to one does not imply the others.

Comparison	What is compared	Evidence available here
Customer value	Equivalent competent human-only work, using conventional software, versus automated delivery	Executed machine resource costs; human duration and residual attention are explicit scenarios, not a timed trial.
Competitive value	Offering versus strongest practical software or AI-assisted alternative	Independent retained Python workflow; pinned converted-oasdiff supplement; buyer options in the simulation.
Investment value	Same enterprise before and after acquiring the capability	Measured A/B/C/D cost deltas, complete hypothetical acquisition costs, finite-life gate. No recurring reconstruction charge.
Network value	Independent agents versus shared registry versus graph-guided reuse	Registry and graph receive identical skills, evidence and dependencies; capability recipients retain their original programs.

4.1 Freeze, confirmation and acceptance

The production manifest froze 61 files at 21:09:19 UTC, before fresh inputs were acquired. It binds source, acceptance, protocols, fixtures, dependencies and construction evidence. The six primary version pairs and two transfer pairs were fixed separately. Each primary arm ran once in rotating order, with a 60-second process bound. Acquisition failures, timeouts and rejected outputs were not replaced or repaired into the primary result. The driver checks source and input identities immediately before production and verification, then checks custody again at the end.

Acceptance independently re-enumerates the supported schema facts and checks every changed tuple, old/new node counts, source locations, reference provenance, input digests and output reconstruction. Missing, extra or duplicate differences fail. Workbench packs are inspected through their real interface. The original conventional workflow has its own separately authored breadth-first verifier. Same-session independence limits shared implementation mistakes; it is not external certification.

The primary sequence is historical public work, fresh to construction. It is not six current customers, six independent organizations or a population estimate. Release lineage is correlated. Only one timing is retained per arm and case; minor timing differences cannot establish a general performance ordering.

Table 4: Primary executed service resources. Includes production, independent verification and packaging; shared retrieval is reported separately.

Arm	Accepted / planned	Total CPU s	CPU s / accepted
A: Retained conventional	3 / 6	1.274743	0.424914
B: Workbench full	3 / 6	4.658206	1.552735
C: Incremental registry	3 / 6	5.304753	1.768251
D: Incremental graph	3 / 6	4.863663	1.621221

The complete primary campaign used 16.139765 CPU seconds and 15.012749 elapsed seconds. Its driver overhead was 0.038400 CPU seconds, allocated separately from the table. The all-input acquisition session used 4.806423 CPU seconds and 7.387453 elapsed seconds. Source availability, normalized bytes, licenses and failed URLs are preserved in the input manifest.

All four arms agree on the exact changed fact tuples and source provenance. The conventional builder is faster in every available primary question. Incremental reuse does not improve the primary average over the full Workbench index. The graph mean is slightly below the registry mean in this single run, but it creates no additional accepted work and supplies no independently identified planning advantage. Both remain slower than the conventional builder.

The public-tool supplement strengthens the alternative. After validated Swagger2-to-OpenAPI conversion, oasdiff 1.34.1 reports 102, 5 and 0 breaking findings and 365, 9 and 77 changelog items on the three available pairs. Its broader request/response/media coverage is not the same acceptance task as the structural dossier. Raw Swagger2 supplied to the OpenAPI3 validation contract produced 15, 1 and 0 findings despite 103–105 validation errors; default exit zero therefore did not establish a valid result. A YAML parser failure on v20 was preserved, then a separately labeled follow-up used the existing frozen normalized JSON. All 27 supplemental commands, including invalid and failed attempts, used 6.280 seconds wall and 8.923 CPU seconds. No primary producer was rerun.

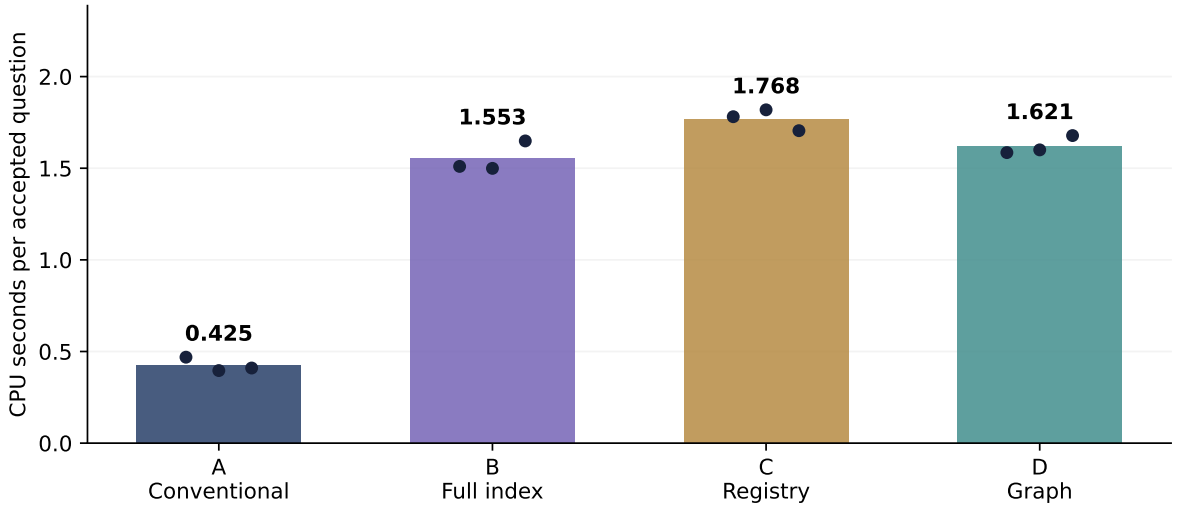


Figure 1: Each dot is one accepted primary question; bars are their mean. Complete compared delivery includes independent verification and packaging. Three blocked questions per arm remain in the intent-to-deliver denominator but do not become zero-duration successful jobs. One pass per case does not establish a latency distribution. Shared acquisition and research costs are separate.

4.2 Transfer to competent recipients

Recipient one built the conventional Python workflow before viewing donor code, after public-tool search. Its observed construction interval was 390.812 seconds. Recipient two independently built another Python implementation in 348.622 seconds. These intervals include tools, reasoning, tests, waits and some context switching; they are neither human task times nor inference invoices. Recipient two integrated the donor module in 44.568 seconds, with 0.956 CPU seconds in the recorded integration commands. This supports only a hypothetical possibility of avoiding initial work for a future recipient. Both tested recipients had already paid the reconstruction effort; no such saving was realized here. It does not justify charging reconstruction on every future job.

Table 5: Original and separately declared exploratory transfer. No original failure is replaced.

Question	Frozen system / recipient one	Independently built recipient two
v23.0.0 → v23.0.6	Four arms pass, but normalized inputs are byte-identical. No changed-work evidence.	Own, received full and incremental methods agree on 2,341 nodes and zero changes. Execution CPU: 0.115, 0.245 and 0.262 seconds before shared validation.
v27.0.0 → v27.5.1	Locked source acquisition fails; all four arms remain unexecuted.	All three methods remain unexecuted.
Exploratory v27.5.1 → v28.0.0	Four narrow-scope outputs pass with 332 changes. CPU: A 0.594, B 2.797, C 2.201, D 1.920 seconds, including verification/packaging.	Existing raw-Swagger validator rejects a required/properties relationship. All three efficacy attempts remain NOT RUN; the validation failure costs 0.664 CPU seconds.

The exploratory follow-up was separately frozen and posted before its outputs, selected only because both already acquired inputs were available and had different byte identities. It used unchanged implementations and acceptance. Its 7.545 CPU seconds and 7.106 elapsed seconds are new research

execution; already paid acquisition is not charged again to the historical book. Browser QA overlapped part of the parent run, limiting interpretation of its wall timing. No performance rerun or input replacement was made, and no secondary result enters the primary market calibration.

Recipient two's selected validator rejects a required `UnpackedSize` member that is not declared under `properties`. Read-only standards adjudication found that this particular check is stricter than the normative relationship: OpenAPI 2 adopts Draft 4's `required` unchanged, which constrains instance membership rather than requiring a schema-property declaration [5, 6]. That does not adjudicate whole-document validity. The original validator HOLD remains authoritative for this recipient's frozen procedure. The difference is an input-qualification boundary, not evidence that the donor completed a broader task or that this recipient received a useful new capability.

The recipient designs differed: one used an independent retained constructor plus breadth-first verification; the other built its own response inventory and then a thin donor wrapper. Both already had effective capability before transfer. The surviving tag-held-out minor-version case has byte-identical normalized specifications and is a content-identity control; the second declared transfer case is acquisition-blocked. Those original tests do not establish transfer to meaningfully changed work. A separately registered one-pair exploratory follow-up uses different-content inputs without changing either implementation; its results are reported separately and excluded from economic calibration. Neither recipient was actually incapable, so this study does not establish valuable positive transfer to such a recipient or skill scarcity.

5 Economic conditions, accounting and investment

Actual collections are zero. Actual complete inference, construction, review, support and maintenance bills are unavailable. The economic study therefore uses executed resource measurements as inputs to an explicitly assumed buyer-and-competitor model. It does not infer an invoice from agent wall time, a price from a desired savings ratio, or demand from the number of generated rows.

The sourced wage anchor is the BLS May 2025 median annual wage of $135,980 for software developers [14]. Dividing by an assumed 2,080 annual hours gives $65.375 per hour. That is a US wage modeling convention, not a Quebec quotation, a fully loaded professional rate or a measured productive hour. Exact-job human durations, customer willingness to pay, integration and residual attention were not observed. Their assumed ranges are declared before simulated outcomes.

5.1 Separate cost books

1. **Marginal execution:** production, independent verification, packaging, failed attempts, source checking and acquisition resources. Internal receipts are not added to outer process-tree CPU again.
2. **Recurring operation:** buyer and provider onboarding, residual attention, support, business overhead, corrections, tool costs and maintenance. These are modeled when not executed.
3. **Full prospective delivery:** recurring costs plus allocated ordinary setup. Experimental four-arm sharing is removed when costing a standalone enterprise.
4. **Incremental capability investment:** construction, failed construction, integration, validation, inference allowance, maintenance and useful-life limits, charged once on acquisition.
5. **Historical research:** discovery and failed discovery, kept separate from prospective deployment and historical recovery. Earlier studies' scenario costs are preserved as prior evidence and are not silently charged again as current invoices.

The root acquisition session fetched or reused each registered source once. Experimental allocations share that work across version appearances and four arms; a standalone policy bears its own service share

rather than receiving a four-arm discount. The companion retains both interpretations and the small driver overhead. Output rows, structural changes and seven Jobs drafts are not additional customers.

5.2 Frozen buyer-and-competitor experiment

The source and calibrated inputs were frozen and posted before the first economic run. There are 24 conditions, from three buyer arrival probabilities (0.025, 0.125 and 0.25 per buyer-week), two buyer integration conditions (0 and 15 minutes), two competitor conditions (equal and stronger), and two full-cost bundles (low and reference). Sixteen paired seeds produce 384 worlds; each contains four separate counterfactual policy histories over 52 weeks. The 24-buyer pool is an assumed simulation population, not an estimated addressable market.

Each enterprise starts with $1,000 seed capital and a $200 investment reserve. The reserve limits discretionary capability investment; it does not prevent ordinary obligations from exhausting cash. Customer payments arrive two weeks after accepted delivery, with a modeled 0.98 collection probability. Unpaid expenses remain obligations. Receivables, seed and internal transfers are not operating-generated investment funds. All book money uses integer cents; investment benefit comparisons retain microdollar precision and conservative cash-rounding effects.

The common 0.5 source-availability condition reflects the six locked questions' 3/6 observed availability, solely as a scenario calibration. A blocked source incurs acquisition and five assumed triage minutes, no delivery retry, onboarding, support or invoice. Available jobs use the observed 3/3 conditional acceptance plus a 0.98 execution-stress assumption and at most two attempts. These are scenario mechanisms, not estimated population probabilities. Human-only/public-DIY delivery uses the same assumed 5, 15 or 45 minutes with competent conventional software. No extra undocumented speed multiplier makes that baseline artificially weak.

The enterprise and equal AI-assisted competitor use the same prospective cost function and tariff rule. The tariff is the smallest member of a labor-equivalent grid that covers expected collected-job cost at three intended orders per week. That cost includes source rejection, execution, correction, support, allocated onboarding, weekly business/maintenance/tools, ordinary setup and compute. A neutral paired draw resolves equal paid offers. The stronger competitor has an explicit 0.6 labor-efficiency assumption. At week 20 competitors can copy, reduce effort and cut prices subject to their prospective cost floor. Buyers can self-serve or decline; orders, delivered jobs, customer avoided labor and collected revenue remain separate.

The low bundle assumes two residual provider minutes per attempt, 0.5 support minutes per order, five provider-onboarding minutes, five weekly business minutes, two weekly maintenance minutes and $1 weekly tools. The reference bundle uses five, two, fifteen, fifteen and five minutes, respectively, and $5 weekly tools. Their modeled historical discovery/setup openings cost $124.42 and $502.63. These labor and inference allocations are not observed invoices. Candidate construction performed in the research remains an unknown actual historical cost; the counterfactual enterprise pays the candidate's modeled incremental cost once only if it purchases that asset. Adding omitted nonnegative historical research costs cannot improve the negative recovery results.

All policies begin with retained conventional A. Policy A makes no purchase; B permits A to B; C permits A to B to C; D permits A to B to D, with graph costs itemized in its second asset. The second purchase cannot bypass its first prerequisite. Investments have a 26-week useful life, an eight-week lookback and earliest investment week eight. Forecasts discount observed reuse and are bounded by remaining useful life. These operating policies are distinct from the separately reported prebuilt A/B/C/D engineering comparison.

Before outcomes, synthetic and independent review corrected asymmetric tariff treatment, unmatched DIY time, acquisition sharing, source-blocked costs and cash rounding. The initial drafts and failed signed-zero control are retained. No simulated profit was used to tune these changes. All 26 final synthetic controls passed; there was one primary market run and no favorable substitute campaign.

5.3 The ordinary service fails before the investment loop

No world produces positive ordinary operating surplus. Across all 384 worlds, available earned investment funds remain exactly zero in every week. No first or second capability is acquired, and no succession repays. The four policy results are identical because every policy retains A. This is an operating-demand/cost failure before a capability funding event, not a claim that graph execution caused the losses.

Table 6: All 24 primary economic conditions, 16 seeds each. Mean accepted jobs and USD results. Policies A/B/C/D coincide because no purchase is admitted. I is buyer integration minutes.

Demand	I	Competitor	Cost	Jobs	Operating	Project
low	0	strong	low	0.00	-448.76	-573.18
low	0	strong	reference	0.00	-509.20	-1011.83
low	0	equal	low	1.00	-447.18	-571.60
low	0	equal	reference	0.00	-509.20	-1011.83
low	15	strong	low	0.00	-448.76	-573.18
low	15	strong	reference	0.00	-509.20	-1011.83
low	15	equal	low	0.81	-445.54	-569.96
low	15	equal	reference	0.00	-509.20	-1011.83
reference	0	strong	low	0.00	-448.76	-573.18
reference	0	strong	reference	0.00	-509.20	-1011.83
reference	0	equal	low	4.25	-439.42	-563.84
reference	0	equal	reference	0.00	-509.20	-1011.83
reference	15	strong	low	0.00	-448.76	-573.18
reference	15	strong	reference	0.00	-509.20	-1011.83
reference	15	equal	low	4.19	-430.30	-554.72
reference	15	equal	reference	0.00	-509.20	-1011.83
high	0	strong	low	0.00	-448.76	-573.18
high	0	strong	reference	0.00	-509.20	-1011.83
high	0	equal	low	8.75	-410.80	-535.22
high	0	equal	reference	0.00	-509.20	-1011.83
high	15	strong	low	0.00	-448.76	-573.18
high	15	strong	reference	0.00	-509.20	-1011.83
high	15	equal	low	9.75	-391.55	-515.97
high	15	equal	reference	0.00	-509.20	-1011.83

The most favorable cell mean is high arrival, 15-minute buyer integration, equal competition and low operating cost: 9.75 accepted jobs, $153.28 simulated collections, $391.55 operating loss and $515.97 modeled project loss. This is a post-run descriptive maximum, not a newly selected confirmatory condition. Even the best individual world loses $233.97 operating and $358.39 after opening costs: $408.75 collected versus $642.72 accrued operating costs and $124.42 opening costs. Its cash is $655.70, unpaid obligations $14.09 and receivables $32.70. Positive cash includes remaining seed and is not profit.

All reference-cost conditions exhaust cash and become insolvent at week 20. Their 19 charged weeks accrue $509.20 operating cost, in addition to $502.63 opening cost, leaving $11.83 unpaid. All low-cost/strong-competitor conditions collect nothing and lose $448.76 in recurring operation, or $573.18 including opening costs. The model makes refusal, competition and finite capital consequential rather than replacing missing sales with assumed transfers.

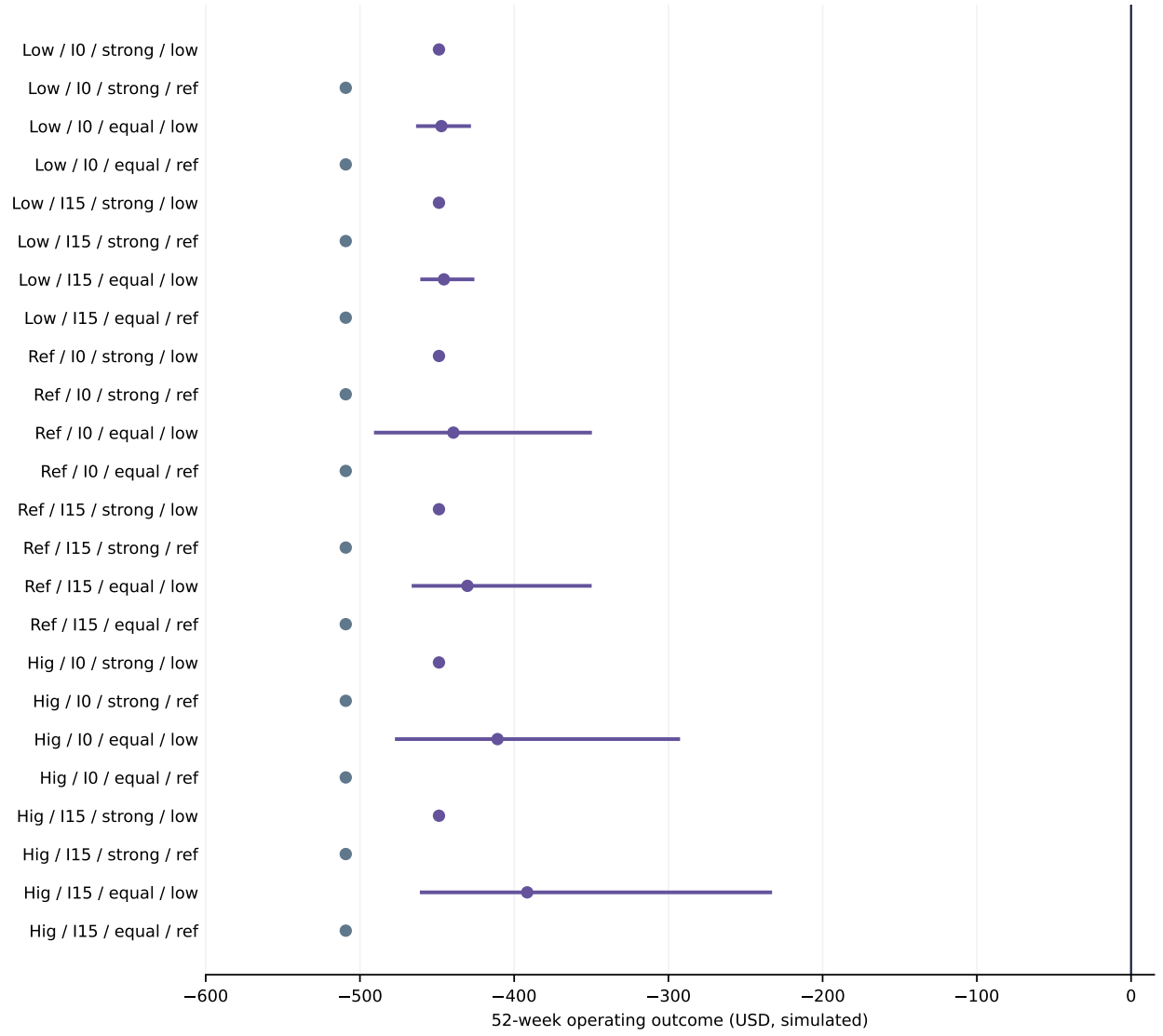


Figure 2: All predeclared cells. Dots show mean operating outcome across 16 paired seeds; lines show the full minimum–maximum range, not confidence intervals. Labels show demand, integration, competition and cost bundle. Every range remains below zero. Reference-cost runs stop after insolvency; they do not receive an extra seed injection.

5.4 Cheap computation is not a complete service margin

Including acquisition, failed source handling at the machine level and driver allocation, the standalone mechanical CPU per accepted question is 1.547, 2.675, 2.891 and 2.744 seconds for A/B/C/D. At the *assumed* $1 per CPU-hour, these are far below a cent. The actual billed machine and inference costs are not known. Residual attention, low conversion and recurring operation dominate the declared scenarios.

Table 7: Prospective costing is not realized utilization. All dollar values below are modeled.

Quantity	Low bundle	Reference bundle
Expected full prospective cost per collected job at reference utilization	$15.48	$35.10
Resulting fixed grid tariff	$16.35	$65.38
Historical discovery/setup opening	$124.42	$502.63
First capability A to B, complete fixed incremental cost	$135.77	$449.95
Second capability B to C, complete fixed incremental cost	$119.43	$363.24
Second capability B to D, including graph increment	$165.53	$509.46

The best cell's actual modeled utilization produces approximately $55.88 accrued recurring cost per accepted job and $68.64 including its opening allocations (ratios of cell totals). Those figures differ from the $15.48 prospective cost because competition and refusal prevent the assumed reference utilization from being achieved. The grid's reference tariff jumps from $35.10 calculated cost to $65.38, above the maximum modeled willingness to pay. This coarse pricing choice contributes to refusal; it is a limitation of the tested policy, not empirical evidence that a continuously priced offering cannot sell. No alternative price campaign was introduced after observing these outcomes.

Human-only duration remains unmeasured. For example, the assumed 45-minute competent workflow costs $49.04 at the wage anchor; a $16.35 service plus two buyer-review minutes costs $18.53 before buyer integration. That conditional $30.51 avoided cost is not observed savings, enterprise revenue beyond the tariff, or available surplus. Shorter assumed human workflows can rationally favor self-service. The diagnostic table varies residual attention over 2, 5 and 10 minutes and retains the unknown actual full-cost category.

5.5 The proposed upgrades also lack a repayment case

A to B adds 1.128 measured CPU seconds per available accepted question; B to C adds 0.216; B to D adds 0.068. These signed deltas supply no positive CPU-only saving even at zero incremental maintenance. The complete candidate investments therefore have no finite CPU-only break-even reuse or repayment time. Adding the declared maintenance cannot improve that conclusion. None was financed.

The separate C-to-D graph diagnostic has a 0.147-second observed CPU difference in the favorable direction, without an added accepted job or identified planning gain. Even if that single-run difference were stable and credited at $1 per CPU-hour, the low $46.10 graph increment would require over 1.15 million reuses at zero maintenance with the frozen conservative microdollar credit (about 1.159 million under the conservative cash-rounding diagnostic). Nonzero declared maintenance removes the finite repayment. This is a sensitivity calculation, not evidence that the measured timing difference generalizes.

At an illustrative 156 execution reuses in 26 weeks, the low-cost A-to-B investment would need about 0.90 minutes of additional paid attention avoided per reuse to cover its complete cost and maintenance. No such saving was observed. This is a concrete measurement target for a future capability test, not a benefit inserted into the present simulation.

5.6 What an investment must repay

Let I be the complete incremental acquisition and validation cost, Δc the signed per-reuse execution saving, m maintenance cost allocated per execution reuse, and N_L feasible execution reuses during useful lifetime L. A necessary condition for repayment is

$$N_L(\Delta c - m) \;\geq\; I, \qquad \Delta c - m > 0.$$

Cash availability is a separate condition. Collected operating funds must cover I after unpaid obligations and the reserve; seed principal and uncollected receivables are not available earned funds. If net saving is nonpositive, no finite break-even reuse or repayment time exists. If a positive timing difference is tiny, its break-even count must remain large rather than be rounded into an attractive per-job margin.

No extra demand, price, attention or reliability premium is credited to the new index or graph without evidence. The graph receives the same catalogue as the registry. A copied public method can still save integration work, but its value must be measured for a recipient that actually needs it. A future improvement that reduces documented review minutes could dominate the CPU effect; this study did not measure such a reduction.

6 Authenticated admission without invented authority

The additive skills admission layer reuses the existing Merkle Roots v114 and PROOF LOOPS v2 foundations through pinned prior authority, commitment and canonical ledger source. The fixtures separate creator identity, eligible `*.agent.agi.eth` validation delegates, root-writing delegates and execution permission. Existing `*.club.agi.eth` Jobs validation remains unchanged.

Records bind implementation versions, declared scope, evidence, dependencies, composition validation, environment, rights, freshness, revocation and supersession. Reuse checks those identities and current status. The full and incremental compositions receive independent technical replay. The final fixture edition binds the final 20-control source-bound review; the previous admission edition and the cost of rebuilding it remain preserved.

All authentication is labeled **FIXTURE ONLY**. Fixture time is not UTC; these identities are not claims that a live eligible ENS agent authenticated this study. Workbench history imports remain unauthenticated historical evidence, active proof and authority are empty, admission is HOLD and settlement is NOT CONNECTED. Merkle inclusion proves only a commitment to supplied bytes. Identity, inclusion and an internally valid ledger do not prove technical correctness, buyer demand or economic value.

The applicable `$AGIALPHA` utility and existing settlement domains remain separate from operating earnings. No live token utility, token transfer or settlement was exercised. Existing USDC and AGIALPHA unit boundaries are not converted into revenue. The model uses USD collections, with no assumed token appreciation or internal-transfer profit.

7 Relation to the continuing research

This work preserves three relevant negative results. *The Next Upgrade* found a strong conventional comparator and no confirmed fresh consumer impact, with complete monetary cost unknown. *The Second Investment* accepted the same six lock questions in all arms while the retained composition was slower; its 1,920 scenario worlds did not qualify a second investment. The *Economy of Verified*

Capability study accepted all 72 scoped evaluations without graph gain and reported no qualifying second investment in 1,152 modeled worlds. Those scenario counts and costs describe those studies, not the present market campaign [15].

The active Capability Advantage lane owns its numerical forward, inverse and constrained-design implementation. It was not duplicated or claimed here. Bounded issue reads did not reveal a separate published checkpoint for Where Autonomy Earns Its Advantage; this absence is not an ownership waiver. Coordination and exact source references are retained in the evidence. Our contribution is the separate portfolio, actual API/GTFS work, competent-recipient tests, source-availability failures and complete-service accounting.

8 Limitations and the concrete missing condition

The biggest limitation is commercial evidence. No buyer commissioned the packet, no willingness to pay was elicited, no paid demand was established and no collections occurred. The research delivered actual scoped outcomes, not an operated commercial enterprise. Human time, inference bills and downstream attention remain unknown. A simulation can illuminate conditions; it cannot turn those unknowns into observed savings or surplus.

The technical limitations also matter. API response coverage excludes requests, live server behavior, full composition semantics, unsupported schema keywords and real consumer execution. The Docker documentation mismatch demonstrates that boundary. The GTFS snapshot's declared calendar covers the requested dates but does not certify real-time operations, actual timetable accuracy or publisher freshness beyond the archive. The source acquisition design failed on two locked URLs and was not quietly repaired after outcomes.

The graph has only two retained skills and one dependency. This is a useful small negative control against unnecessary architecture, not a theorem about larger heterogeneous networks. One primary pass per arm does not estimate stable latency distributions. The originally available transfer case is content-identical; its separately registered changed-content follow-up is exploratory. Both recipients were already capable, and historical releases are correlated. These facts prevent claims of scarce general capability or profitable transfer.

The missing condition is specific: a recurring buyer job for which an independently assessed, competently equipped recipient still faces a costly gap, and a retained improvement that measurably reduces the complete accepted-delivery burden enough to repay acquisition, integration, validation and maintenance during its useful life. The next justified experiment would begin with that buyer's acceptance contract and timed incumbent workflow, not with a more elaborate graph. A reliable input resolver or source-plus-runtime consumer verifier may be candidates, but neither is financed or validated as a successor by this study.

9 Conclusion

The work produced usable API review dossiers, a checked transit ingestion packet, reusable source-bound capabilities, fresh negative-control transfer evidence and a finite-cash model. It also preserved blocked inputs, strong conventional alternatives and a genuine schema-versus-runtime scope failure. The tested assets do not establish scarce capability or a uniquely advantageous construction method. The economic conclusion is reported at its demonstrated level in the results, with no conversion of simulated receipts into actual revenue.

The productive discipline is to keep useful delivery, reject unjustified investments, and retain the evidence needed to choose a better next job. Failing to close this bounded cycle does not establish that the broader AGI ALPHA opportunity is absent.

A Reproduction and evidence map

The code and recorded evidence are under `studies/agi-alpha-work-discovery-v1/`. The companion retains the repository-relative Workbench dependency closure so that its imports resolve without changing the research source. Node 24.22.1, Python 3.12.14/PyYAML 6.0.3 and Java 17 were used. Public tool versions and licenses are retained. Runtime installations are prerequisites, not claimed to be embedded operating systems.

Evidence group	Location relative to the study
Preselection and primary freeze	`protocol/discovery.json`, `production.json`, `production-freeze.json`
Complete primary execution	`outputs/primary/campaign.json`; each case retains command streams, source-bound service output and acceptance
Acquisition and licensing	`inputs/acquired/manifest.json`; originals, normalized JSON, per-version licenses and failed URLs
Competent recipients	`comparators/conventional/`, `comparators/recipient/`; original source freezes, integration and once-only cost records
Original transfer and successor	`outputs/transfer/`, `outputs/secondary-transfer/`, `protocol/secondary-transfer-freeze.json`
Public alternative	`evidence/oasdiff-comparison.md`, raw/converted outputs and the separate normalized-input follow-up
GTFS complete delivery	`discovery/gtfs/output/`, `discovery/gtfs/workbench/`, original archive, license, validator and SQLite receipts
Actual consumer boundary	`discovery/consumer-impact/`; authentic source, license, unmodified extracted-function probe and EN/FR advisory
Admission and prior work	`src/admission/`, `heritage/graph-foundations/`, `heritage/prior-studies/`; originals and fixture editions retained
Finite-cash model	`src/economics/`, `protocol/economics-freeze.json`, `outputs/economics-primary/`; calibration, protocol, every weekly history and summaries
Presentation	`demo/index.html`, `paper/main.tex`, generated result fragments and scientific figures

For a new authorized pair of normalized specification files, run the retained conventional constructor and its independent verifier, or the actual Workbench adapter:

```
python comparators/conventional/conventional.py OLD.json NEW.json OUT/dossier.json
python comparators/conventional/verify_dossier.py OLD.json NEW.json OUT/dossier.json
node src/run_service.mjs OLD.json NEW.json NEW_OUTPUT admitted-full
python tests/accept_dossier.py OLD.json NEW.json NEW_OUTPUT
```

Run from the study directory. Output directories must be new. Ordinary consumers should inspect the declared supported scope and warnings before using a dossier for a decision. Guarded Workbench modes use local fixtures and do not authenticate a live agent.

Research replay must occur in a disposable copy. The original outcome directories are immutable. The primary driver accepts the exact published freeze hash; it refuses source drift, existing campaign outputs or changed input manifests. Replaying timings is a reproduction attempt, not another primary sample. The economic runner likewise requires its exact source and calibration freeze and a new output directory. Do not regenerate the archived data in place.

Publication is an additive research proposal, separate from a deployed service or main-release qualification. Required repository merge and release checks retain their existing meaning. No frozen native HOLD test or protected execution path was opened by this study.

B English/French demonstration

The companion's offline demonstration uses one embedded result object for both languages. It presents complete jobs, comparisons and transfer, and finite-cash scenarios. Unknown costs remain unknown. It makes no network request, asks for no account, and performs no deployment, transaction or settlement.

Démonstration en français

Le travail livré comprend des dossiers de modification d'API et un dossier de validation structurelle du flux GTFS du REM. Les résultats sont rattachés aux sources et aux vérifications exécutées. Les trois paires d'API disponibles ont satisfait les contrôles dans les quatre approches; trois autres paires restent bloquées par l'acquisition des sources. Le programme conventionnel conservé consomme moins de processeur que les compositions testées.

Les deux agents destinataires avaient déjà reconstruit une capacité équivalente avant de recevoir le module. Le cas initial disponible est identique par son contenu. Un essai exploratoire distinct, enregistré séparément, porte sur des contenus différents et conserve les échecs initiaux. Le transfert ne démontre ni capacité rare ni amélioration économique récurrente. L'identification ENS et les attestations sont des exemples explicitement étiquetés; elles ne donnent aucune autorité d'exécution réelle.

Les scénarios économiques séparent le capital initial, les encaissements simulés, les coûts, les obligations et la réserve. Les durées humaines, la demande et la disposition à payer sont des hypothèses, non des observations. Aucun paiement client n'a été encaissé. Une économie simulée favorable pour une prestation ordinaire ne suffit pas à justifier un nouvel investissement: son avantage doit rembourser son coût complet pendant sa durée utile.

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